THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 5: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The smallest number which leaves remainder 5 when divided by 12 and 18 is: 1

(a) 221 **(b)** 41 **(c)** 77 **(d)** 36

2. The number of multiples of 7 between 100 and 500 (both included) is: 1

(a) 56 **(b)** 58 **(c)** 59 **(d)** 57

3. The number of 3-digit numbers divisible by 8 is: 1

(a) 111 **(b)** 112 **(c)** 113 **(d)** 124

4. The smallest 3-digit palindrome divisible by 3 is: 1

(a) 121 **(b)** 101 **(c)** 111 **(d)** 114

5. If the number 6a2b is divisible by 90, then $a + b$ equals: 1

(a) 2 **(b)** 3 **(c)** 4 **(d)** 1

6. The number of steps the Kaprekar routine needs to reach 6174 starting from 2111 is: 1

(a) 5 **(b)** 6 **(c)** 7 **(d)** 4

7. A number is divisible by both 9 and 11. It must be divisible by: 1

(a) 108 **(b)** 20 **(c)** 198 **(d)** 99

8. The smallest number which leaves remainder 7 when divided by 15 and 20 is: 1

(a) 127 **(b)** 67 **(c)** 307 **(d)** 60

9. **Assertion (A):** The number 4,236 is divisible by 3. 1

Reason (R): A number is divisible by 3 if the sum of its digits is divisible by 3.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The number 7,424 is divisible by 4. 1

Reason (R): A number is divisible by 8 if the number formed by its last three digits is divisible by 8.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the smallest number which leaves remainder 3 when divided by 20 and 25. 2

12. How many multiples of 11 lie between 150 and 800 (both included)? 2

13. The sum of five consecutive numbers is 45. Find the numbers. 2

14. Find the smallest number which leaves remainder 5 when divided by 24 and 36. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Show that the six-digit number 247247 is divisible by 7, 11 and 13. 3

16. Find the greatest 4-digit number divisible by 6, 8 and 12. 3

17. How many 3-digit numbers are divisible by both 6 and 8? 3

OR

Show that the six-digit number 358358 is divisible by 7, 11 and 13. 3

18. Find the greatest 4-digit number divisible by 10, 12 and 15. 3

OR

How many 3-digit numbers are divisible by both 9 and 12? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A school gives each student a four-digit locker code. Neha's code is 7,236 and Tarun's code is 52**d**4, where d is a missing digit. 5

(i) Is 7,236 divisible by 9?

(ii) Is 7,236 divisible by 4?

(iii) Is 7,236 divisible by 11?

(iv) Find the digit d so that 52d4 is divisible by 9, and check that the number is then also divisible by 6.

20. Find the digits a and b if the number 6a2b is divisible by 90. List all possibilities. 5

OR

Take the number 641. Subtract its reverse. Add the reverse of the result to the result. What do you get? Show that this always happens for 3-digit numbers whose first and last digits differ by 2 or more. 5

END OF QUESTION PAPER — **Total: 40 Marks**